

Summer School 2026

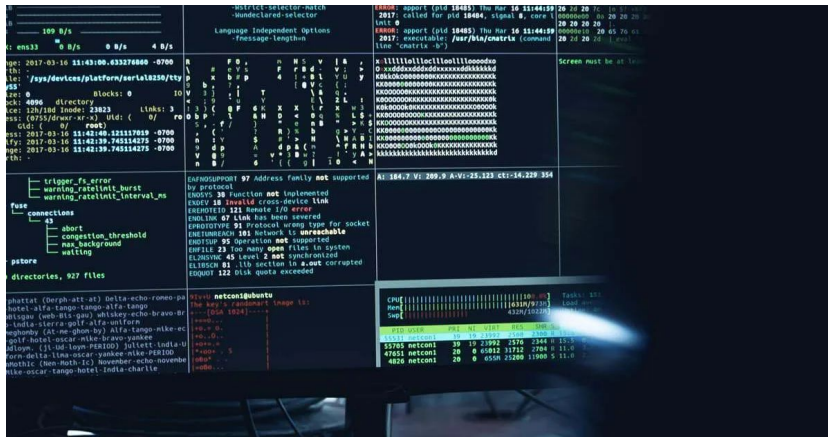
Introduction to Capture The Flag (CTF)

bubu





- **Hacker Nickname**
- **Wear a Black Hoody.**
- **Work during the night or without light.**
- **Green letters in the console.**



Summer School 2026

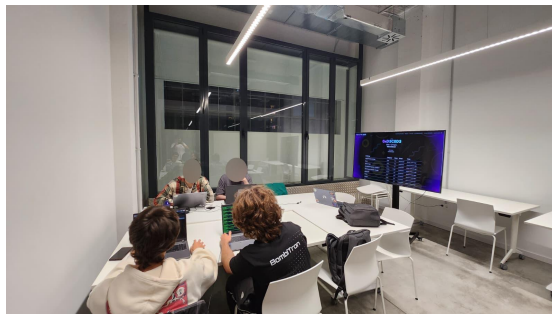
What is a CTF?

Capture The Flag



CTF

///
0x
D



MOLECON 2022

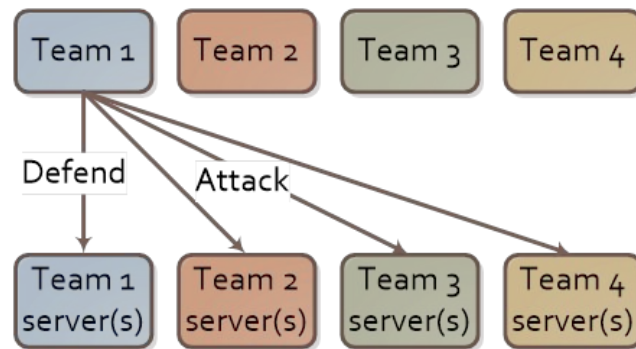


0xdecode{esto_es_una_flag}

Goal → Flag

Jeopardy

Web	Crypto	Forensics	Reverse	Misc
1	165	100	50	50
150	150	150	100	100
204	150	150	150	165
203	200	200	200	150

Attack-Defense



CTF - Jeopardy

Retos individuales divididos en categorías.

- **PWN / Exploiting**
- **Reversing (Rev)**
- **Web Security (Web)**
- **Crypto**
- **Misc**
- (Forense)
- (Hardware)
- (Steganography)

Web	Crypto	Forensics	Reverse	Misc
1	165	100	50	50
150	150	150	100	100
204	150	150	150	165
203	200	200	200	150

- DEFCON CTF quals: the only thing that gets a room of professional research staff looking through 10,000 photos of antique furniture for a weekend

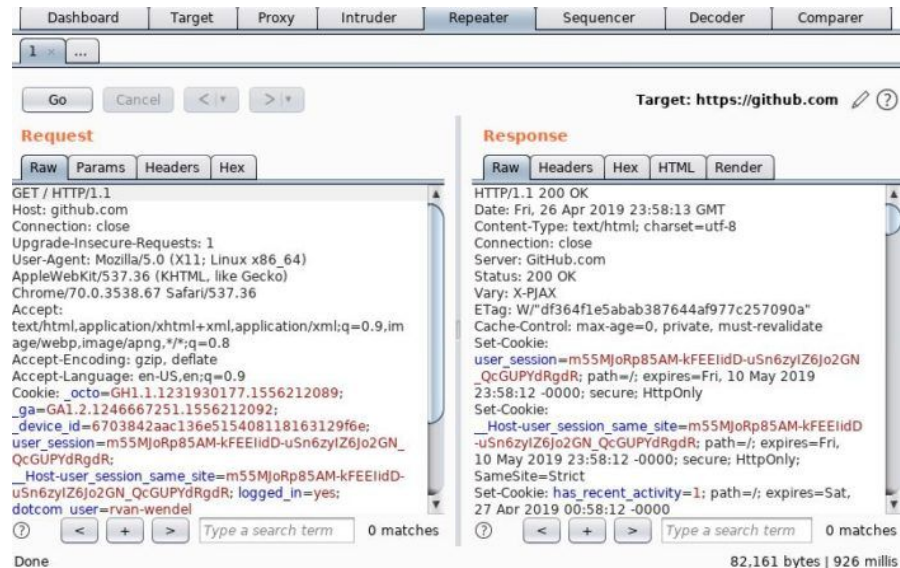


CTF - Jeopardy - Web

Jeopardy - Web Security

- Webpage Exploitation.
 - Server-Side Template Injection.
 - Command Injection.
 - ...

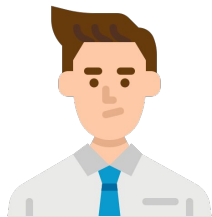
```
~/E/CT/H/Web/WildGooseHunt > python3 solve.py
Found one more char : CHTB{1
Found one more char : CHTB{1_
Found one more char : CHTB{1_t
Found one more char : CHTB{1_th
Found one more char : CHTB{1_th1
Found one more char : CHTB{1_th1n
Found one more char : CHTB{1_think
Found one more char : CHTB{1_think_
Found one more char : CHTB{1_think_t
Found one more char : CHTB{1_think_th
Found one more char : CHTB{1_think_the
Found one more char : CHTB{1_think_the
```



BurpSuite

CTF - Jeopardy - Web

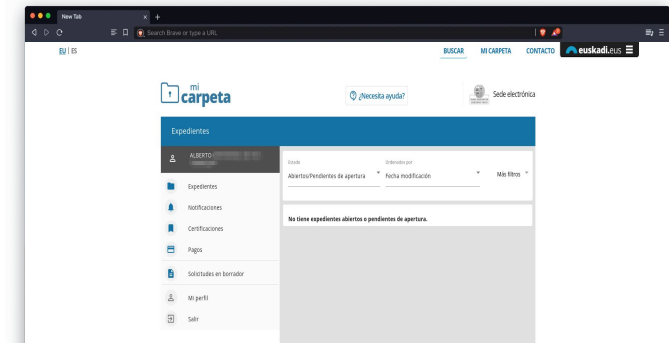
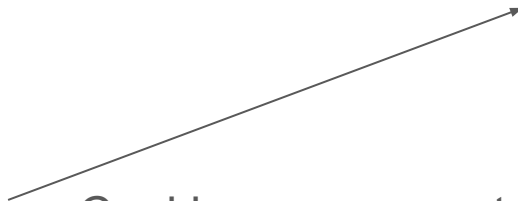
Cross-Site Scripting (XSS)



Cookie: user=secret



Cookie: user=secret

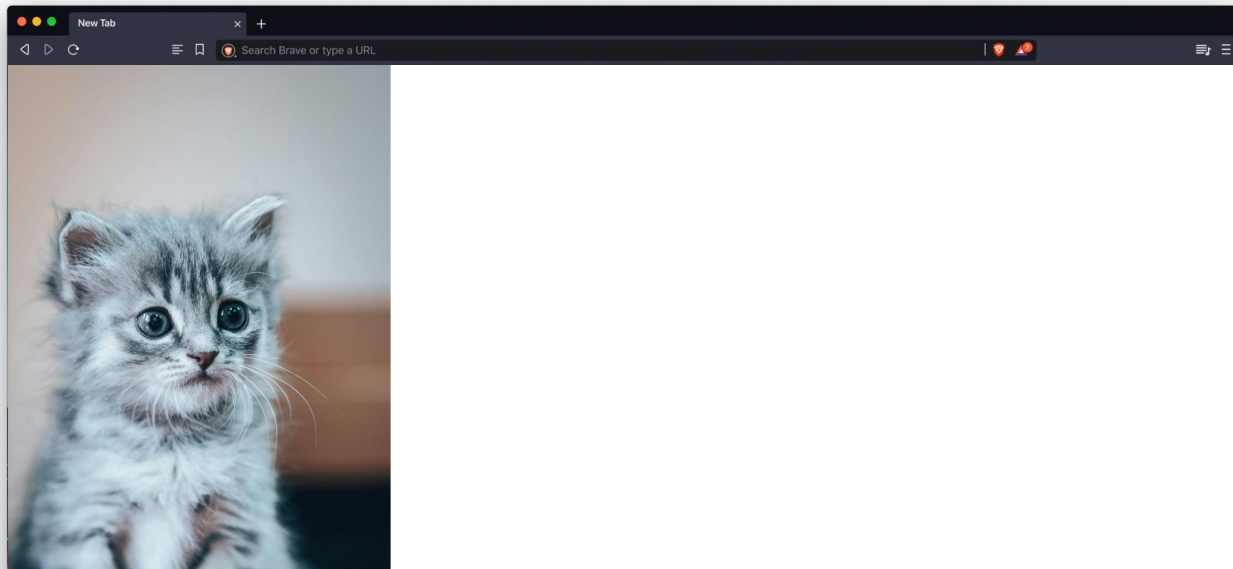




CTF - Jeopardy - Web - XSS

Cross-Site Scripting (XSS)

`https://euskadi.eus/fake.html?text=<img src=gatito.jpg>`

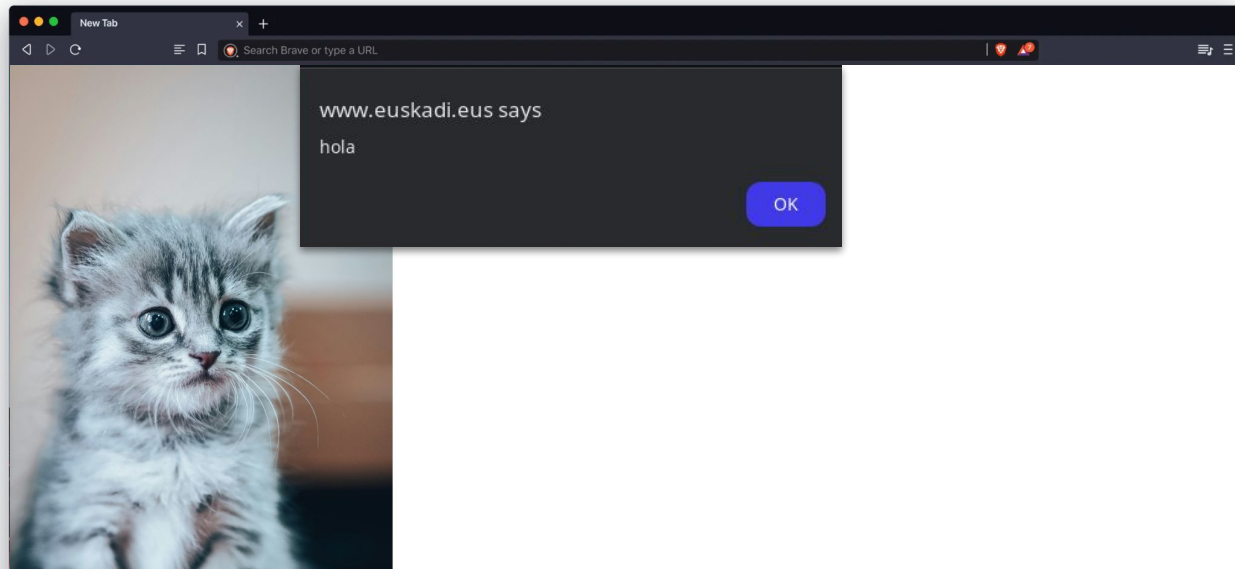




CTF - Jeopardy - Web

Cross-Site Scripting (XSS)

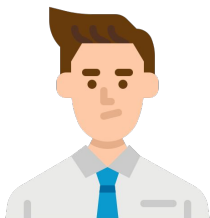
`https://euskadi.eus/fake.html?text=<img src=gatito.jpg onload=alert("hola") >`



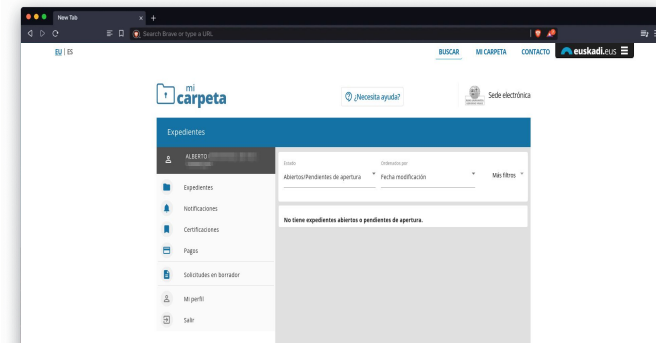
CTF - Jeopardy - Web

Cross-Site Scripting (XSS)

`https://euskadi.eus/fake.html?text=PAYLOAD`



Cookie: user=secret



Cookie: user=secret





CTF - Jeopardy - Web

SQL Injection

```
sql_statement = "SELECT * FROM users WHERE user = " + USER + " AND password = " + PASSWORD + ";"
```

<https://0xdeco.bank/?user=bubu&password=bubu>

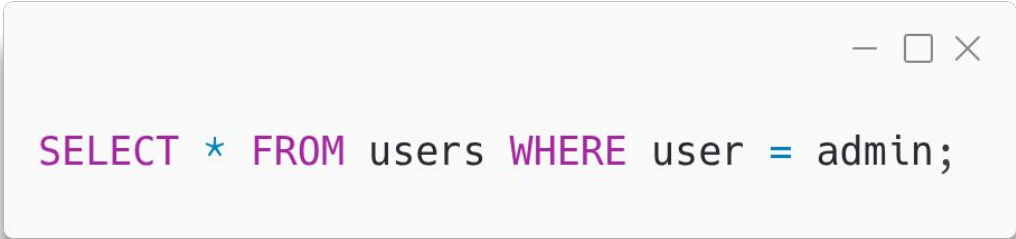
```
SELECT * FROM users WHERE user = bubu AND password = bubu;
```



CTF - Jeopardy - Web

SQL Injection

`https://0xdeco.bank/?user=admin;&password=bubu`



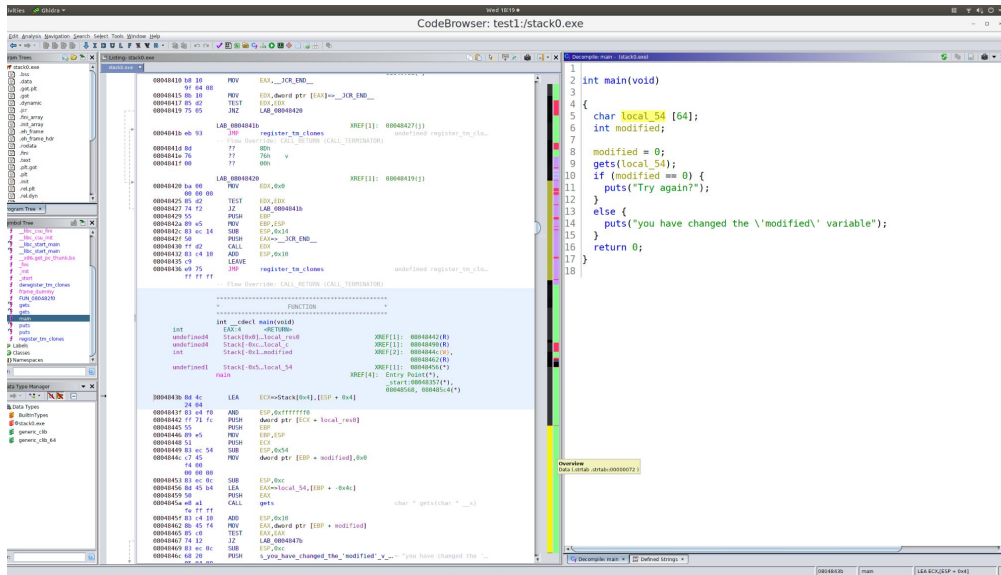
```
SELECT * FROM users WHERE user = admin;
```



CTF - Jeopardy

Reversing

- Binary reversing.
 - Recognize patterns.
 - Search the comparison of the characters of the password.
- Tools like Binary Ninja, IDA, r2, Ghidra o Cutter



```
> python3 exploit.py
=====
SOLUTION:98304

> nc chall.polygl0ts.ch 3600
What's your your number:98304
EPFL{4ft3r_th15_h0w_h4rD_caN_r3V_8e?}
```

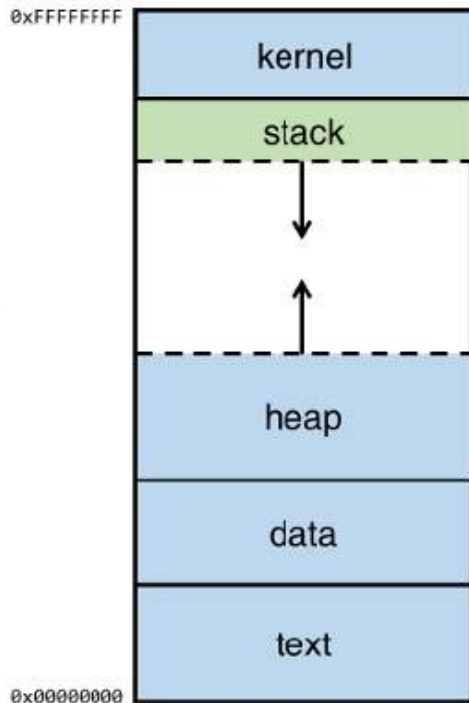
Ghidra



CTF - Jeopardy

PWN/Exploiting

- Binary Exploitation
 - ELF (Linux)
- Find user/attacker inputs that let us execute the commands we want, pwning/exploiting the program.



Buffer overflow

Return address
Caller's frame pointer
[0][1][2][3][4][5][6][7]
Local variables

CTF - Jeopardy

PWN/Exploiting

- Format string vulnerability

```
printf("%s", input);
```

```
input: %p %p %p
resultado: %p %p %p
```

```
printf(input);
```

```
input: %p %p %p
resultado:
0x0
0xdeadbeef
0x7fffffff7a33450
```



CTF - Jeopardy

Crypto (Conceptos básicos)

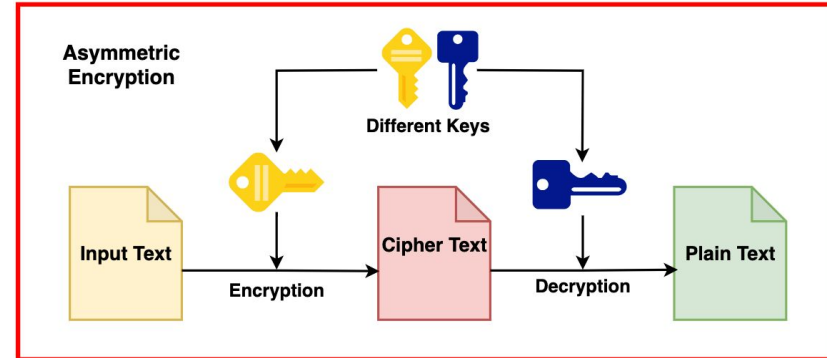
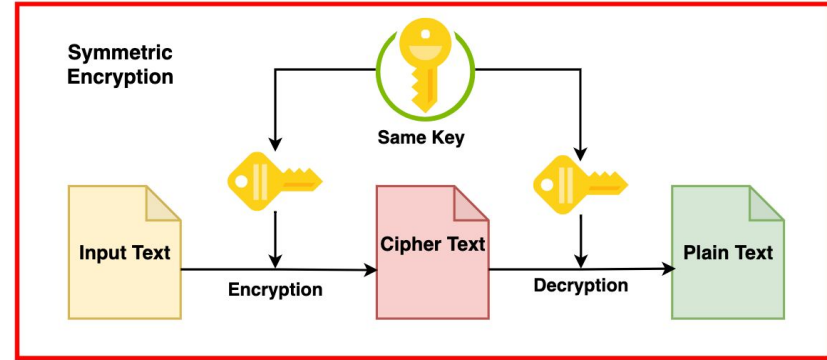
- Hexadecimal
- Base64
- Hash

CTF - Jeopardy



Crypto

- Symmetric Encryption
- Asymmetric Encryption



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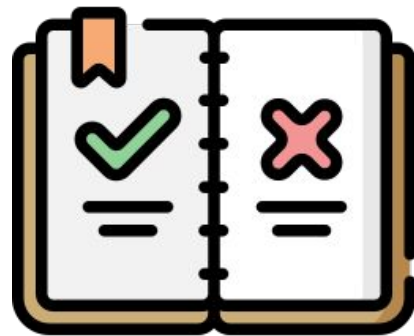
0xDECODE CTF





Rules

- Do not share or post any flags, hints or solutions publicly during the CTF.
- Cross-team collaboration is not allowed. Do not discuss challenges with members from other teams.
- Attacking event infrastructure (DoS / DDoS) is strictly prohibited.
- Do not bruteforce flags.
- **No LLM allowed.**

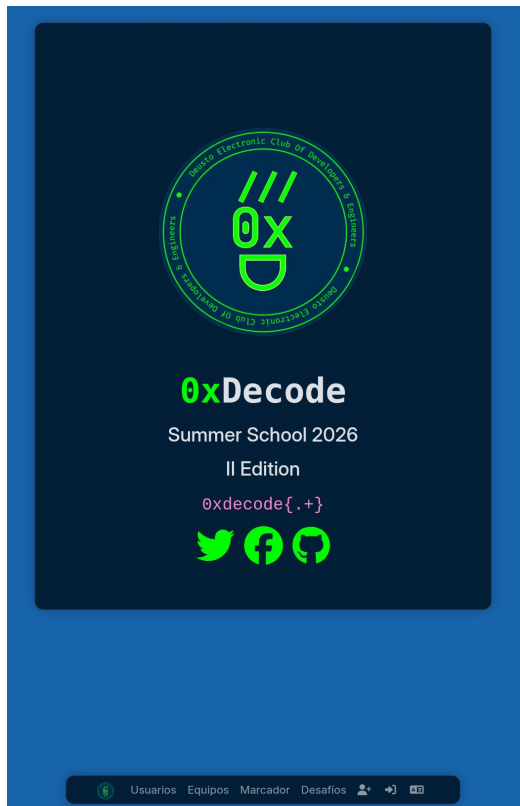




Deusto

Universidad de Deusto
Deustuko Unibertsitatea
University of Deusto

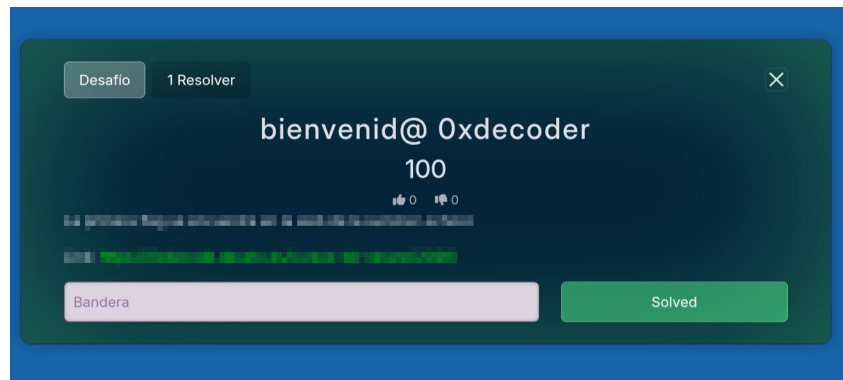
Agradecimiento por el espacio, y
más importante, por la pizza 🍕



0xDECODE CTF



- Duración: Ahora - 17:00
 - En algún momento 🍕
- Equipos de 4 personas
 - Random ..



<https://192.168.1.222/>